Homely

Project Overview:

Homely is a modern e-commerce platform designed to provide users with a seamless and convenient shopping experience for home essentials and everyday household products. The platform allows users to browse a variety of items, explore product details, and manage their shopping cart efficiently.

Homely aims to simplify the process of purchasing home products by offering a clean, user-friendly interface combined with dynamic features that enhance interaction and usability. The system focuses on delivering a responsive design, secure authentication, and smooth data handling to ensure a reliable user experience.

Through its structured design, Homely enables users to easily navigate between products, add items to their cart, and interact with the system in real-time using modern web technologies. The platform is built to be scalable and can be extended in the future with additional features such as order tracking, payment integration, and user reviews.

Key Features:

The Homely platform includes a set of essential features designed to provide a smooth and interactive e-commerce experience:

- **User Authentication System**
  Secure user registration and login functionality to manage user accounts safely.
- **Product Browsing**
  Users can explore a variety of home essential products displayed in an organized and user-friendly layout.
- **Product Details View**
  Each product has a dedicated page showing detailed information such as description, price, and image.
- **Shopping Cart System**
  Users can add, update, and remove products from their cart with dynamic price calculation.
- **AJAX-Based Interactions**
  Real-time updates such as adding items to the cart or updating quantities without reloading the page.
- **Responsive Design**
  The platform adapts seamlessly to different screen sizes including mobile, tablet, and desktop devices.
- **Secure Form Handling**
  All forms include validation and security measures such as CSRF protection to ensure safe data handling.
- **API Integration**
  The system provides API endpoints for handling product and cart data dynamically.

Functional Requirements:

The Homely platform must provide the following core functionalities:

- **User Registration & Login**
  Users can create a new account, log in securely, and log out of the system.
- **Product Listing**
  The system displays all available products in an organized layout for easy browsing.
- **Product Details**
  Users can view detailed information about each product, including description, price, and image.

- **Add to Cart**
  Users can add products to their shopping cart.
- **Update Cart**
  Users can update product quantities or remove items from their cart.
- **Cart Management**
  The system dynamically calculates the total price based on selected items and quantities.
- **AJAX Interaction**
  Users can perform actions such as adding or updating cart items without reloading the page.
- **Navigation Between Pages**
  Users can easily navigate between different pages such as Home, Product Details, Cart, and About Us.
- **User Session Management**
  The system maintains user sessions to keep users logged in and preserve cart data.

Technical Requirements:

The Homely platform will be developed using the following technologies and tools:

- **Backend Framework**
  Django (Python) will be used to handle server-side logic, routing, and data processing.
- **Database**
  MySQL will be used as the relational database to store and manage application data.
- **Frontend Technologies**
  HTML, CSS, and JavaScript will be used to build the user interface.
- **Responsive Design**
  Bootstrap (or CSS media queries) will be used to ensure the application is fully responsive across different devices.
- **AJAX Integration**
  AJAX will be implemented to enable dynamic updates without reloading pages.
- **API Implementation**
  RESTful API endpoints will be created to handle data interactions such as products and cart operations.
- **Authentication & Security**
  Secure authentication will be implemented using Django's built-in system, including password hashing and CSRF protection.
- **Version Control**
  Git and GitHub will be used for version control and project documentation.

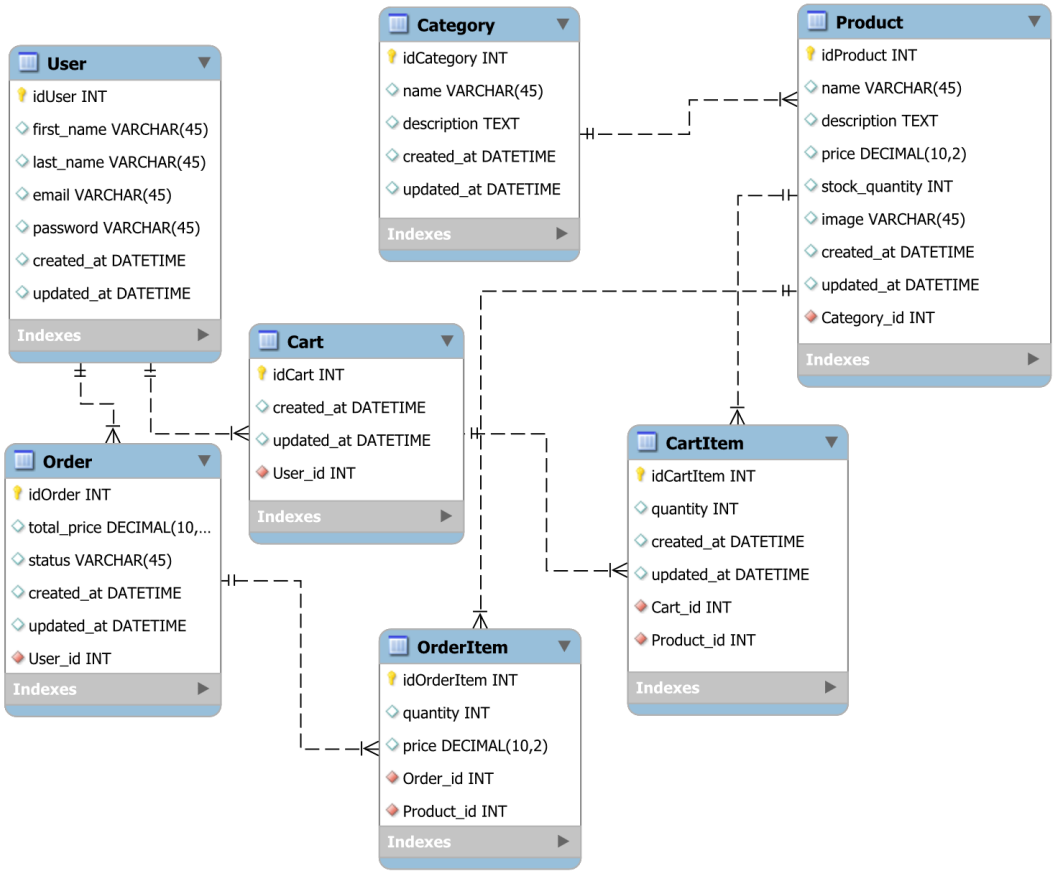
Non-Functional Requirements:

The Homely platform should meet the following quality standards:

- **Usability**
  The system should provide a simple and intuitive interface that is easy for users to navigate.
- **Performance**
  The application should load quickly and handle user interactions efficiently.
- **Reliability**
  The system should operate consistently with minimal errors or downtime.
- **Security**
  The platform must ensure secure handling of user data through authentication, validation, and protection mechanisms.

- **Responsiveness**
  The application should adapt smoothly to different screen sizes and devices.
- **Maintainability**
  The code should be clean, organized, and well-documented to allow easy updates and future improvements.
- **Compatibility**
  The system should work across modern web browsers and devices without issues.

ERD:



ERD Description:

- **User**
  Stores user information such as name, email, and password used for authentication and account management.
- **Category**
  Represents product categories used to organize items in the store.
- **Product**
  Represents items available in the store, including details such as name, price, description, and stock.
- **Cart**
  Represents the shopping cart associated with a specific user.

- **CartItem**
  Represents products added to the cart along with their quantities.
- **Order**
  Represents a completed or pending purchase made by a user.
- **OrderItem**
  Represents individual products within an order, including quantity and price at the time of purchase.

Wireframes:

The following wireframes represent the main pages of the Homely e-commerce platform. These designs illustrate the user flow starting from browsing categories to viewing product details, as well as authentication and cart management.

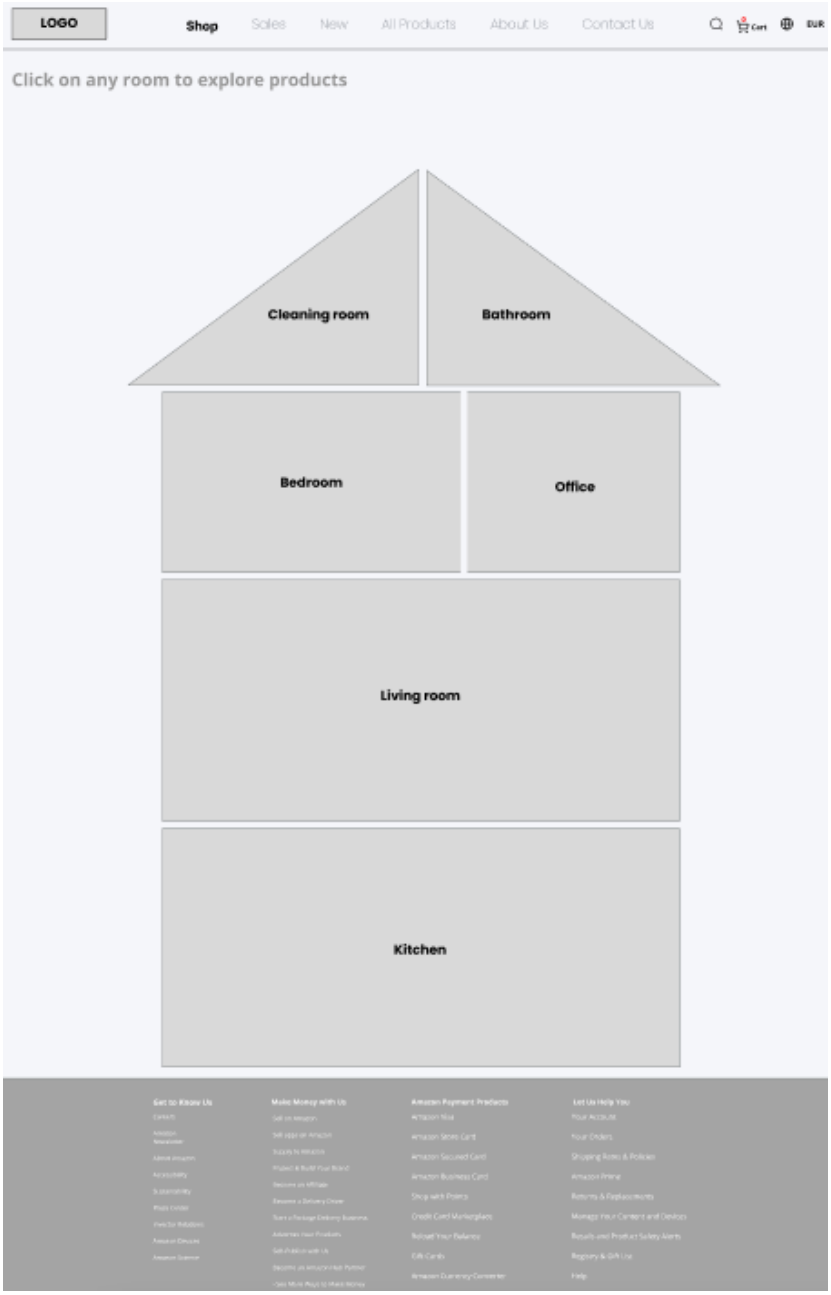


Figure 1: Home Page Wireframe

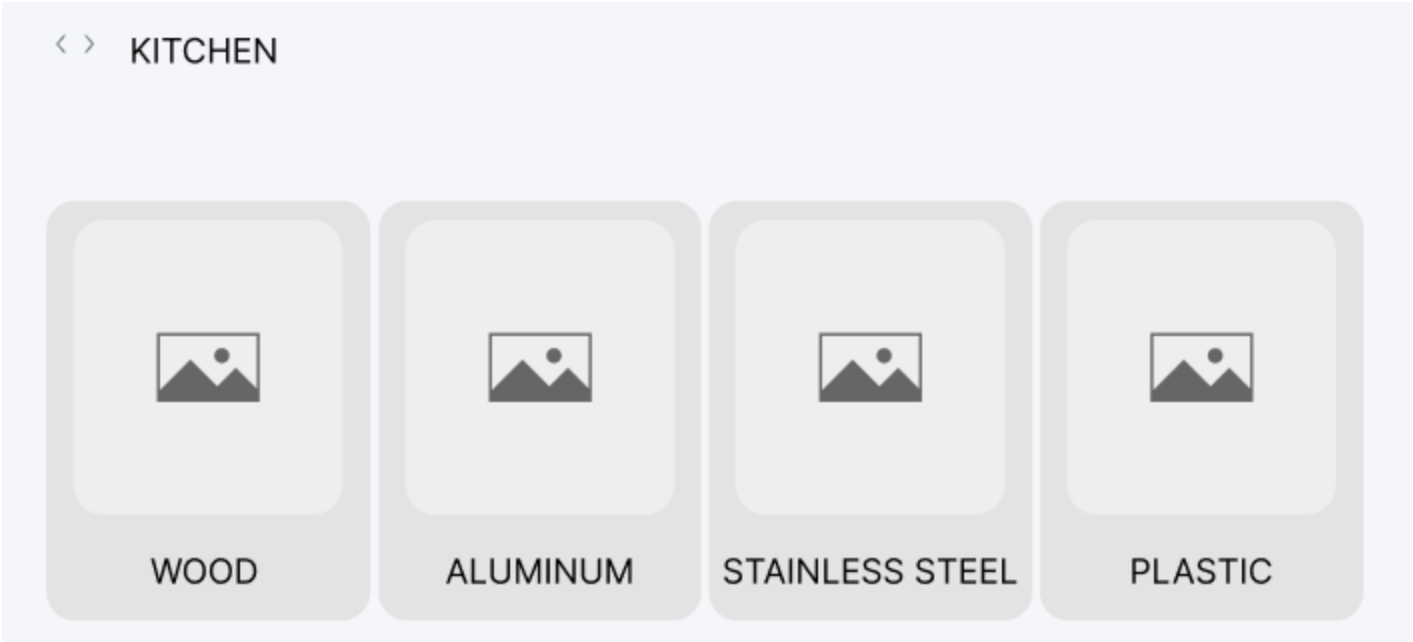


Figure 2 : Category Page (Kitchen)

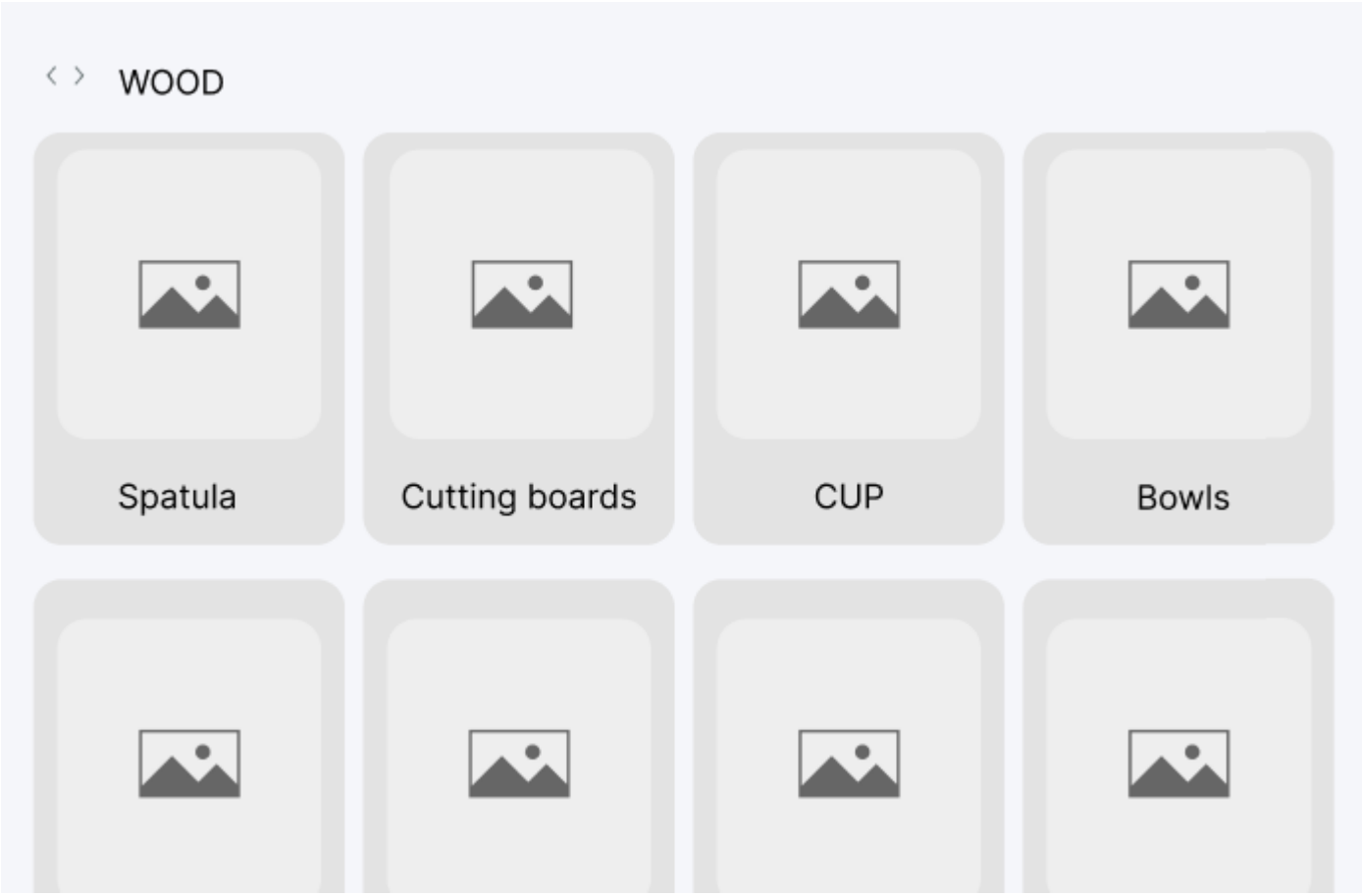


Figure 3 : Product List Page (Wood Category)

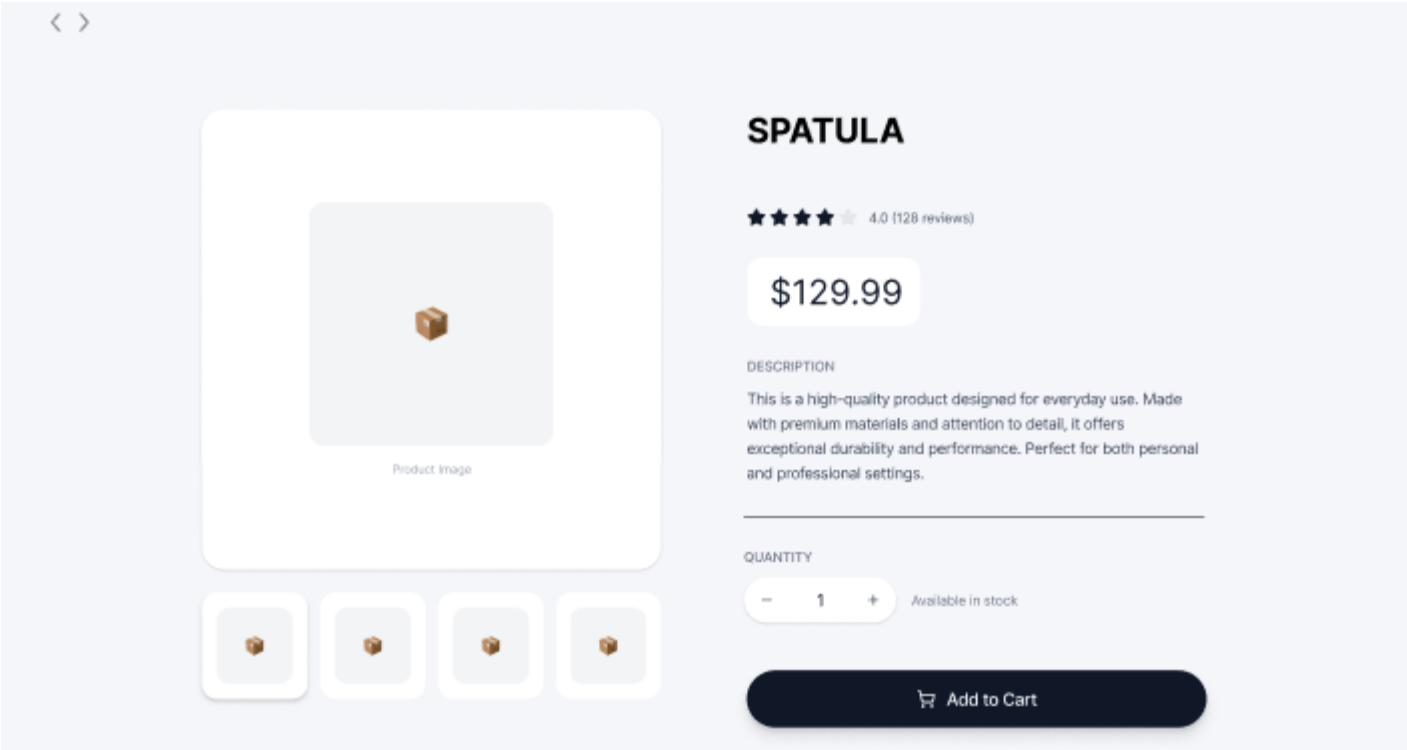


Figure 4 : Product Details Page

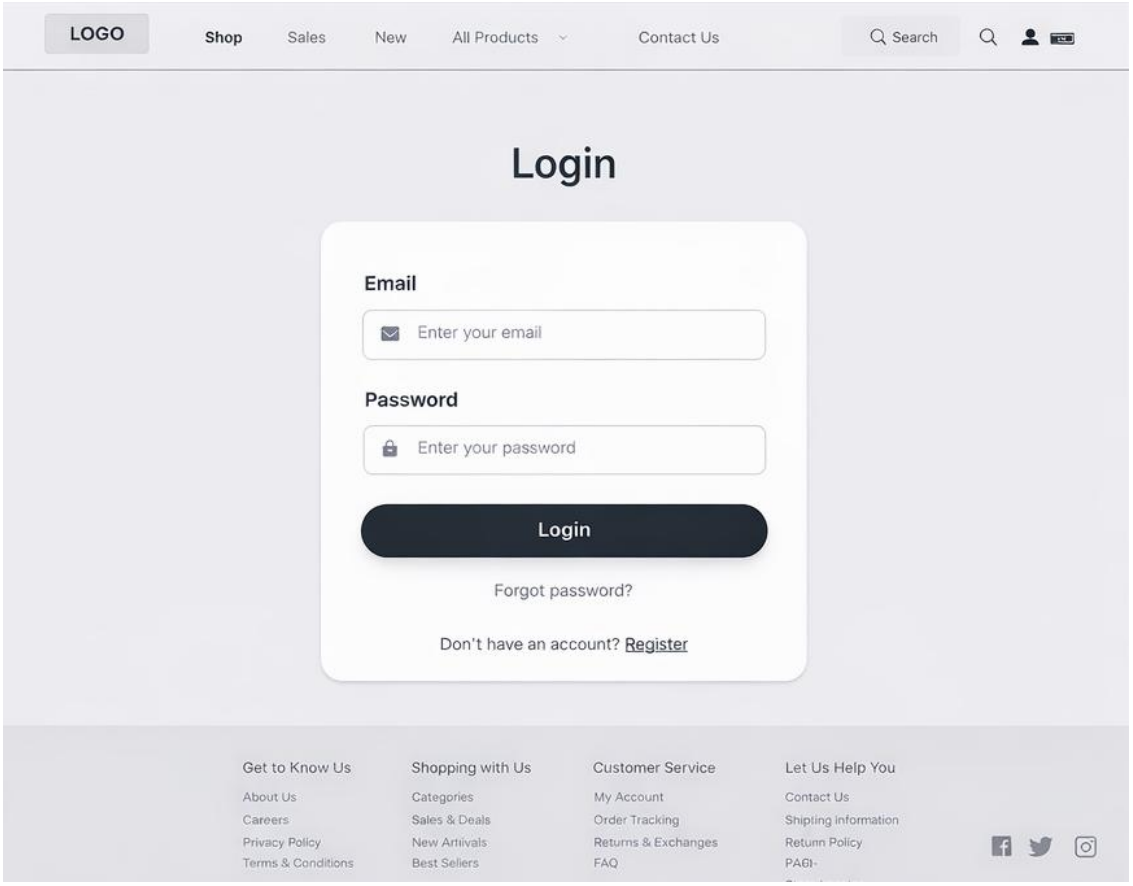


Figure 5 : Login Page

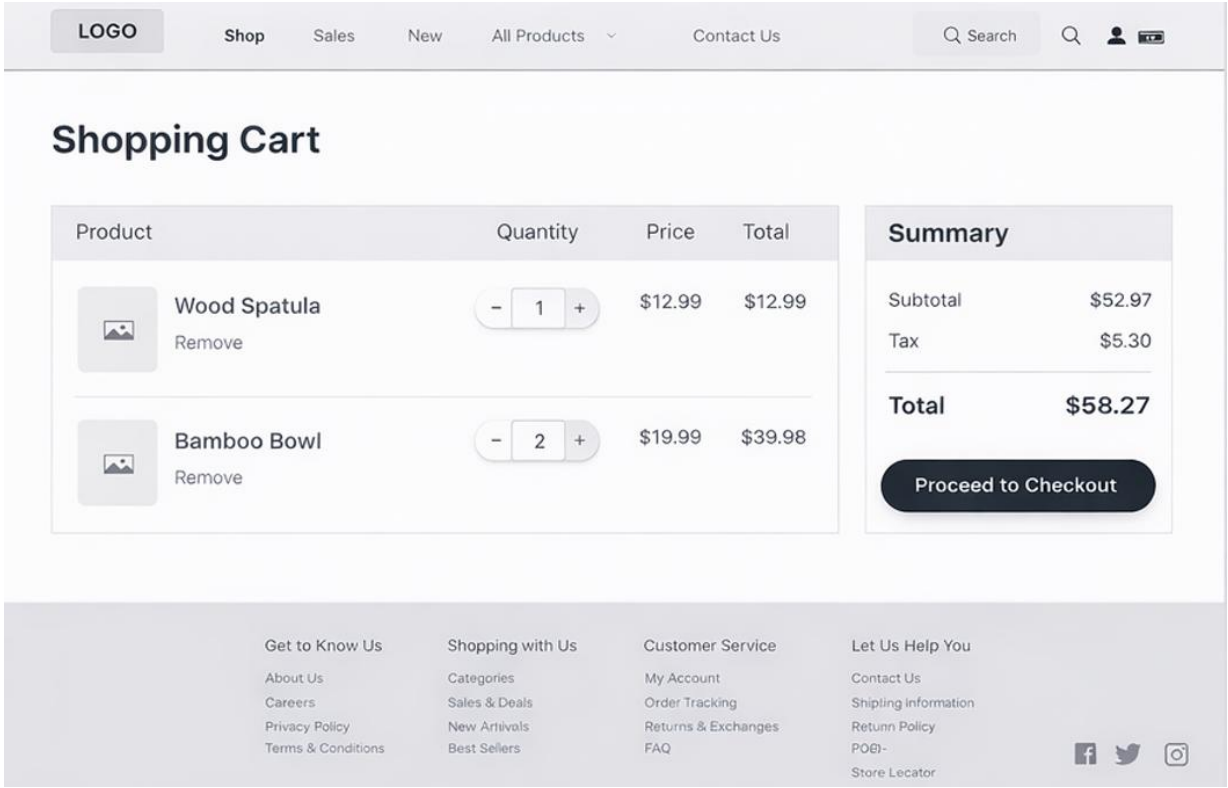


Figure 6 : Shopping Cart Page

Links:

- [Trello]
- [GitHub]